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DT/NT : DT
LESSON : LINUX
**SUBJECT: LINUX FILE PERMISSIONS &
PACKAGE MANAGEMENT**
SESSION : 4
BATCH : B 424-020DE

AWS-DEVOPS



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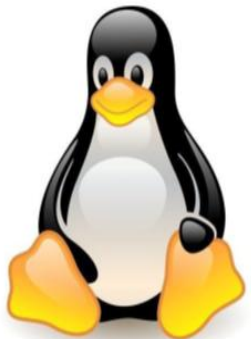


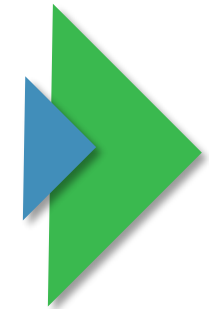
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Files and Directories



Files and Directories



- The file system hierarchy standard (FHS) defines the structure of the file systems on Linux.
- In the FHS, all files and directories appear under the root directory /, even if they are stored on different physical or virtual devices.
- Most of these directories exist in all UNIX, however, they are not considered authoritative for platforms other than Linux.

Files and Directories



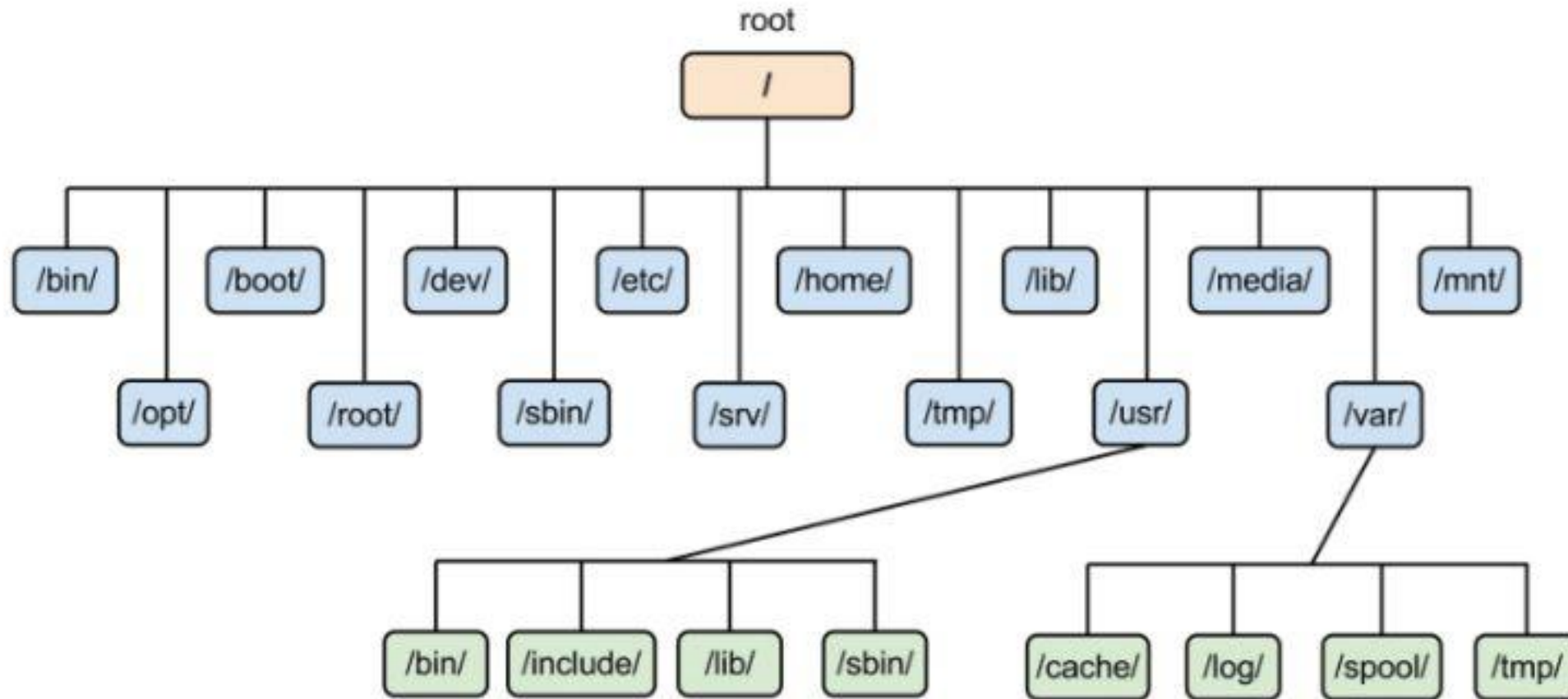
- On a Linux system, everything is a file.
- A Linux system makes no difference between a file and a directory, since a directory is just a file containing names of other files.
- The tree of the file system starts at the trunk or slash, indicated by a forward slash (/). This directory, containing all underlying directories

Files and Directories

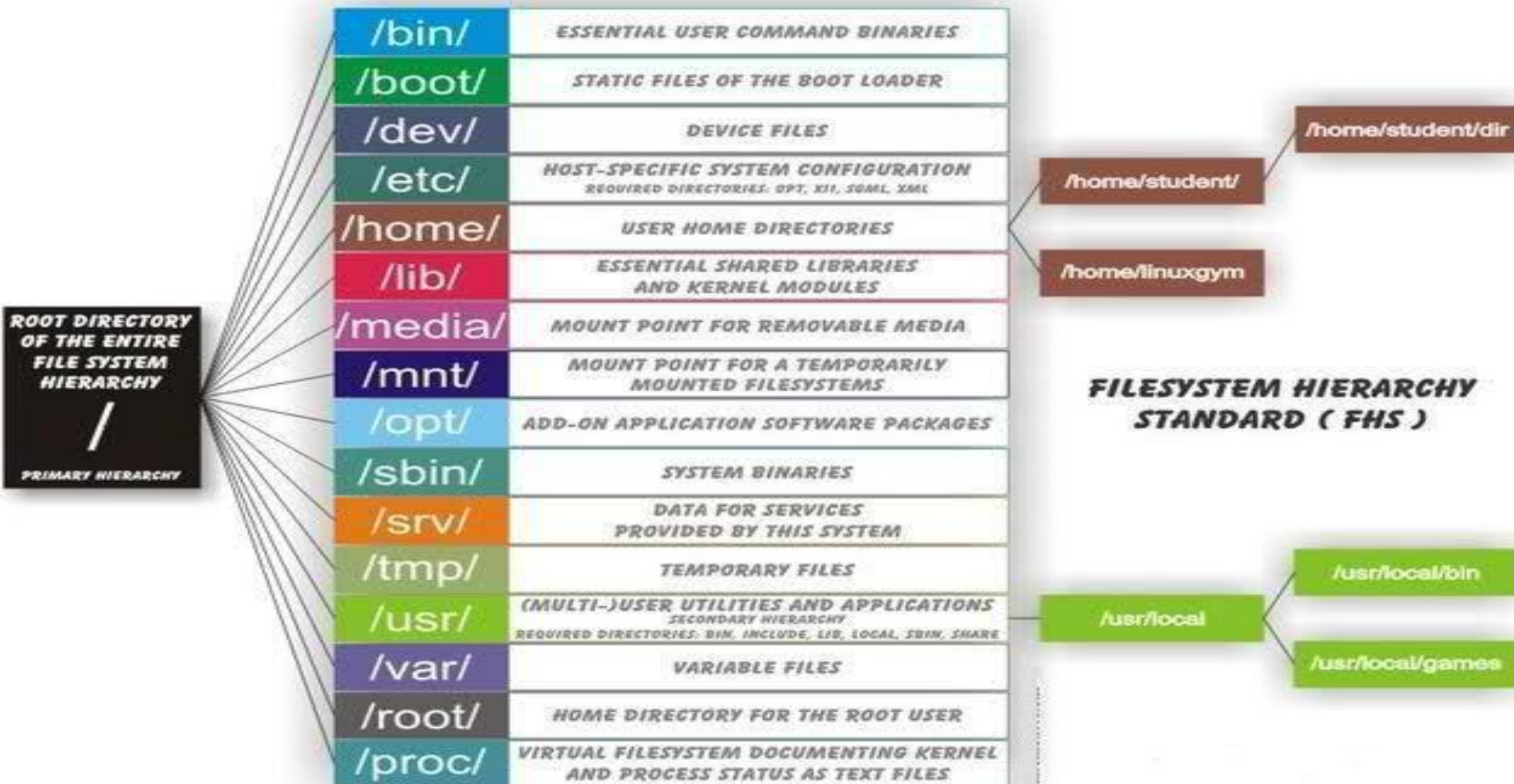


<code>/root</code>	• Home directory of the root user
<code>/bin</code>	• Essential command binaries
<code>/boot</code>	• Boot loader files
<code>/dev</code>	• Essential device files
<code>/etc</code>	• Host-specific configuration files
<code>/home</code>	• Users' home directories
<code>/lib</code>	• Libraries essential for the binaries
<code>/mnt</code>	• Temporarily mounted filesystems.
<code>/opt</code>	• Optional application packages
<code>/proc</code>	• Contains information about system
<code>/sbin</code>	• Essential system binaries
<code>/tmp</code>	• Temporary files
<code>/var</code>	• Variable data files

Files and Directories



Files and Directories



Files and Directories



Symbol	Meaning
-	Regular file
d	Directory
l	Link
c	Character Device File
s	Socket File
p	Named Pipe
b	Block Device

-rw-----	Regular File
d rw-xr-xr-x.	Directory File
l rw-rw-rw-r.	Link File
c rw-rw----.	Character Device File
b rw-rw----.	Block Special File
s rw-rw-rw-	Socket File
p rw-----.	Named Pipe File

Files and Directories

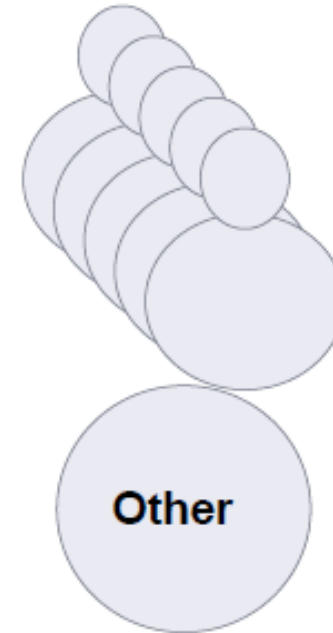
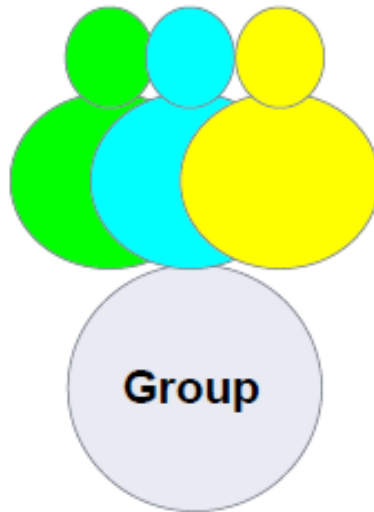
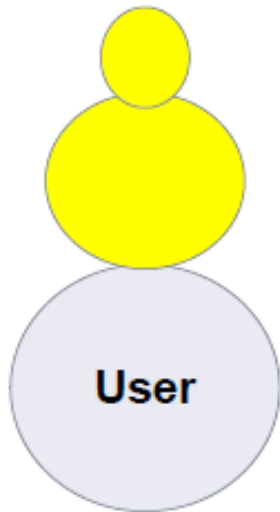


- On most Linux versions ls is aliased to color-ls by default. This feature allows to see the file type without using any options to ls.

Color	Meaning
Blue	directories
Red	compressed archives
White	text files
Pink	images
Cyan	links
Yellow	Devices
Green	Executables
flashing red	broken links

```
linuxuser@techproeducation: X + v
linuxuser@techproeducation:/$ cd lib
linuxuser@techproeducation:/lib$ ls
apparmor          initramfs-tools    recovery-mode
apt               kernel             rsyslog
binfmt.d          locale             sasl2
byobu             lsb                snapd
cnf-update-db     man-db             software-properties
command-not-found mime                ssl
compat-ld         modprobe.d         sysctl.d
console-setup     modules-load.d     systemd
dbus-1.0          netplan            sysusers.d
dpkg              networkd-dispatcher tc
environment.d     openssh             terminfo
file              os-release          tmpfiles.d
girepository-1.0  pam.d              ubuntu-advantage
git-core          pm-utils            ubuntu-release-upgrader
gnupg             policykit-1         udev
gnupg2            polkit-1            ufw
gold-ld           python2.7           usrmerge
groff             python3             valgrind
hdparm            python3.10          wsl
init              python3.11          x86_64-linux-gnu
linuxuser@techproeducation:/lib$ |
```

File Permission



Ownership



File Permission



Read



Write



Execute



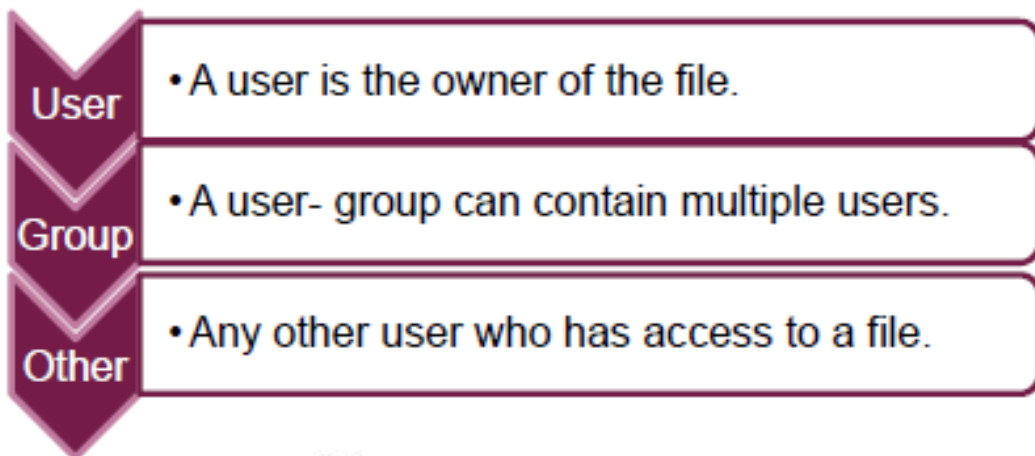
Ownership



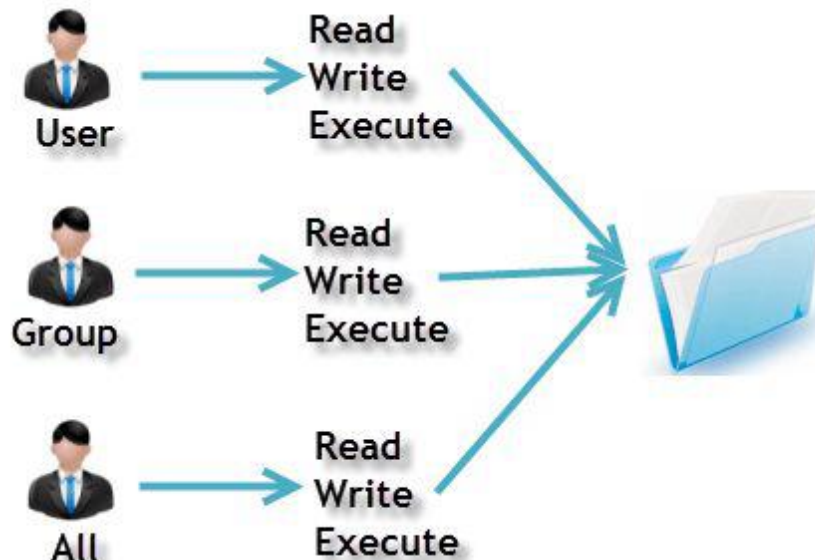
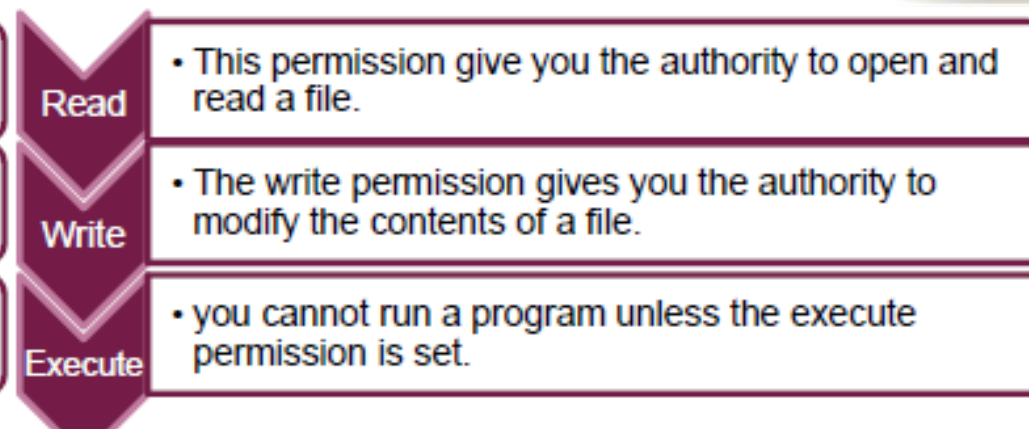
File Permission



Ownership



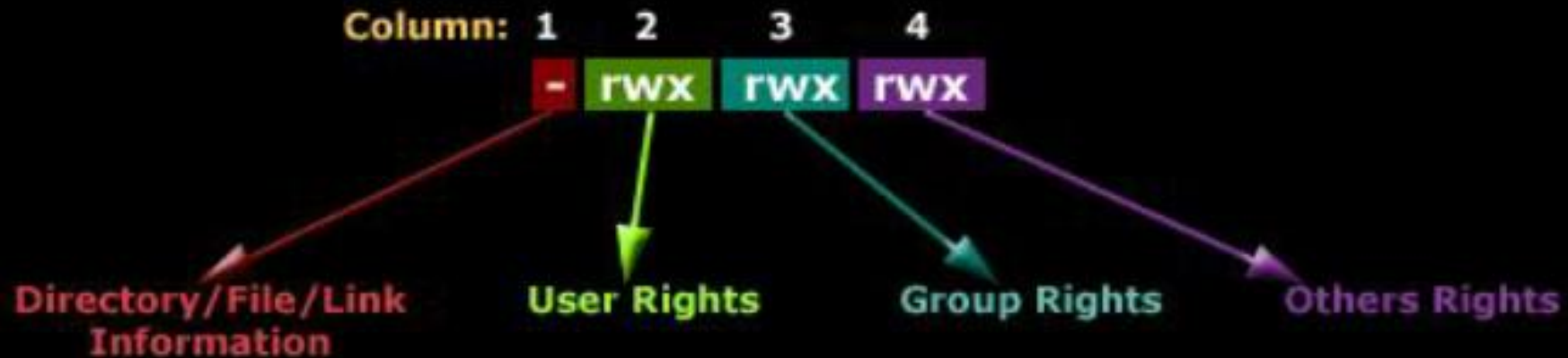
Permission



File Permission

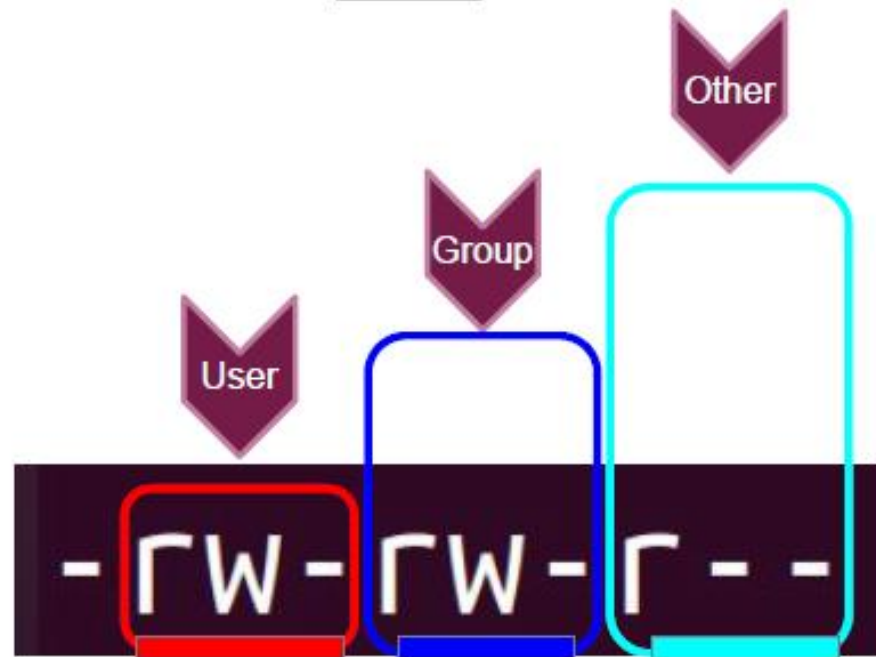


Understanding The Linux File Permissions



While the first column defines a directory, file or link, the next 3 columns (2, 3, 4) define the permissions for the User, Group and Others (everyone else) groups.

File Permission



```
-rw-rw-r-- 1 zk zk 0 Dec 7 15:39 html.txt
```



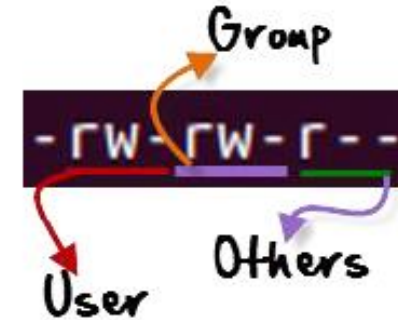
Ownership



File Permission



```
linuxuser@techproeducation: ~$ ls -l
total 21260
-rw-rw-r-- 1 linuxuser linuxuser 21765972 Nov 30 11:03 latest-hugo.zip
drwxrwxr-x 2 linuxuser linuxuser 4096 Nov 30 09:56 linux_lesson
linuxuser@techproeducation: ~$
```



r: Read
w: Write
x: Execute

File type and Access Permissions

-rw-r--r-- 1 gakeko2018 gakeko2018 807 Dec 25 18:19 .profile

indicates File

-rw-rw-r--

no execute permission

r = read permission
w = write permission
x = execute permission
- = no permission

drwxr-xr-x 1 gakeko2018 gakeko2018 4096 Jan 13 09:41 .

d represents directory

File Permission



Changing Permission with chmod Command

- We can use the chmod command which stands for change mode.
- we can set permissions (read, write, execute) on a file/directory for the owner, group and the world.

```
chmod permissions filename
```

```
chmod u=rwx,g=rx,o=r myfile
```

Symbol	Permission Type
---	No Permission
--x	Execute
-w-	Write
-wx	Execute+Write
r--	Read
r-x	Read+Execute
rw-	Read+Write
rwx	Read+Write+Execute

File Permission



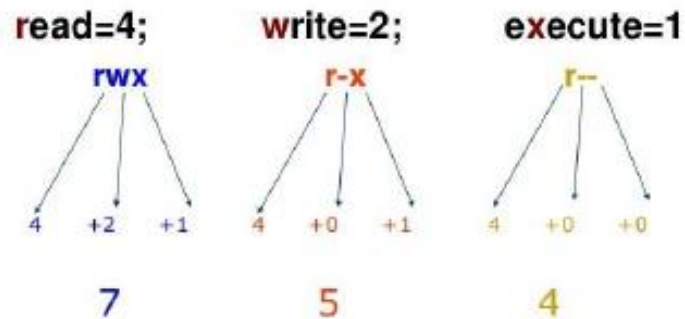
```
linuxuser@techproeducation: ~  
linuxuser@techproeducation:~/permission$ ls -l  
total 0  
-rw-rw-r-- 1 linuxuser linuxuser 0 Dec 1 10:40 myfile  
linuxuser@techproeducation:~/permission$ chmod 754 myfile  
linuxuser@techproeducation:~/permission$ ls -l  
total 0  
-rwxr-xr-- 1 linuxuser linuxuser 0 Dec 1 10:40 myfile  
linuxuser@techproeducation:~/permission$
```

754 code says;

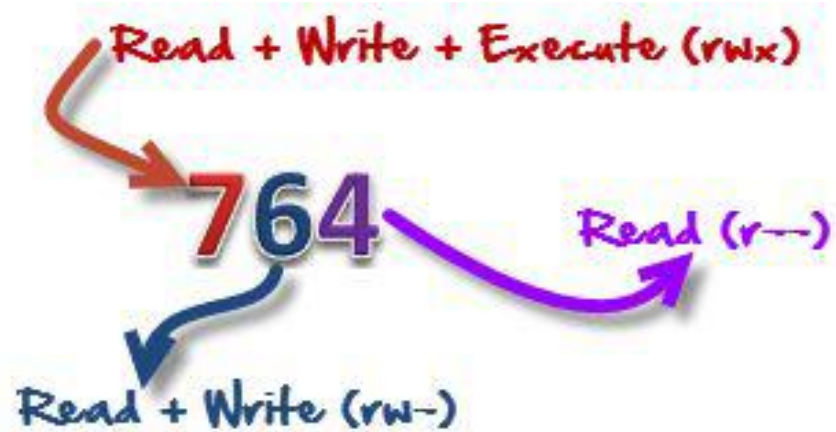
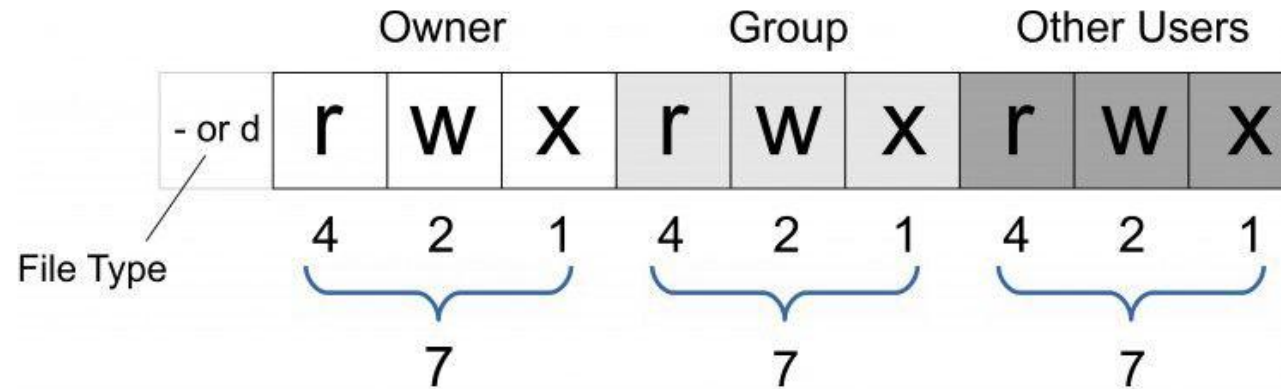
- Owner can read, write and execute
- User's group can read and execute
- Other can only read

```
chmod u=rwx,g=rx,o=r myfile  
chmod 754 myfile
```

Permissions



File Permission



d	r	w	x	r	-	x	r	-	-
	read	write	exec	read	write	exec	read	write	exec
File type	Owner permissions			Group permissions			User permissions		
(directory)	4	2	1	4	2	1	4	2	1
	7			5			4		

File Permission



File Attributes	Meaning
-rwx-----	A regular file that is readable, writable, and executable by the file's owner. No one else has any access.
-rw-----	A regular file that is readable and writable by the file's owner. No one else has any access.
-rw-r--r--	A regular file that is readable and writable by the file's owner. Members of the file's owner group may read the file. The file is world-readable.
-rwxr-xr-x	A regular file that is readable, writable, and executable by the file's owner. The file may be read and executed by everybody else.
-rw-rw----	A regular file that is readable and writable by the file's owner and members of the file's group owner only.
lrwxrwxrwx	A symbolic link. All symbolic links have "dummy" permissions. The real permissions are kept with the actual file pointed to by the symbolic link.
drwxrwx---	A directory. The owner and the members of the owner group may enter the directory and create, rename and remove files within the directory.
drwxr-x---	A directory. The owner may enter the directory and create, rename, and delete files within the directory. Members of the owner group may enter the directory but cannot create, delete, or rename files.

File Permission



Octal	Binary	File Mode
0	000	- - -
1	001	- - X
2	010	- W -
3	011	- W X
4	100	r - -
5	101	r - X
6	110	r W -
7	111	r W X

By using three octal digits, we can set the file mode for the owner, group owner, and world.

File Permission



Notation	Meaning
u+x	Add execute permission for the owner.
u-x	Remove execute permission from the owner.
+x	Add execute permission for the owner, group, and world. This is equivalent to a+x.
o-rw	Remove the read and write permissions from anyone besides the owner and group owner.
go=rw	Set the group owner and anyone besides the owner to have read and write permission. If either the group owner or the world previously had execute permission, it is removed.
u+x, go=rx	Add execute permission for the owner and set the permissions for the group and others to read and execute. Multiple specifications may be separated by commas.

File Permission



Set permissions of myfile.txt to;

owner : full access

group : read and execute

others : no access

File Permissions in DevOps Practice



Correct permissions protect credentials and keep services working.

SSH PRIVATE KEY

```
chmod 400 mykey.pem
```

- Only the owner should be able to read the private key.
- Overly open permissions can cause SSH to reject the key.

APPLICATION LOGS

```
-rw-r----- app.log
```

- Owner: read + write
- Group: read
- Others: no access

DevOps habit: grant only the permissions a user or service actually needs.

Inspecting File Metadata with stat



Use stat when ls -l is not enough.

COMMAND

```
stat app.log
```

Typical information

- Owner and group
- Permission mode
- File size
- Access / modify / change timestamps

TROUBLESHOOTING USE

```
File: app.log  
Access: (0640/-rw-r-----)  
Uid: ec2-user  
Gid: devops
```

Useful when a deployment cannot read, write, or execute a file as expected.

Default Permissions with umask



umask removes permission bits from the default permissions of new files and directories.

```
umask 022
```

NEW FILES

Base 666 → 644

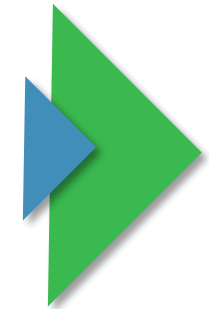
Typical result: `rw-r--r--`

NEW DIRECTORIES

Base 777 → 755

Typical result: `rw-r-xr-x`

Important: umask removes permissions; it does not add them.



Package Management



Package Management



A package manager is a collection of software tools that automates the process of installing, upgrading, configuring, and removing computer programs for a computer's operating system in a consistent manner.



Package Manager



Office Boy

Package Management



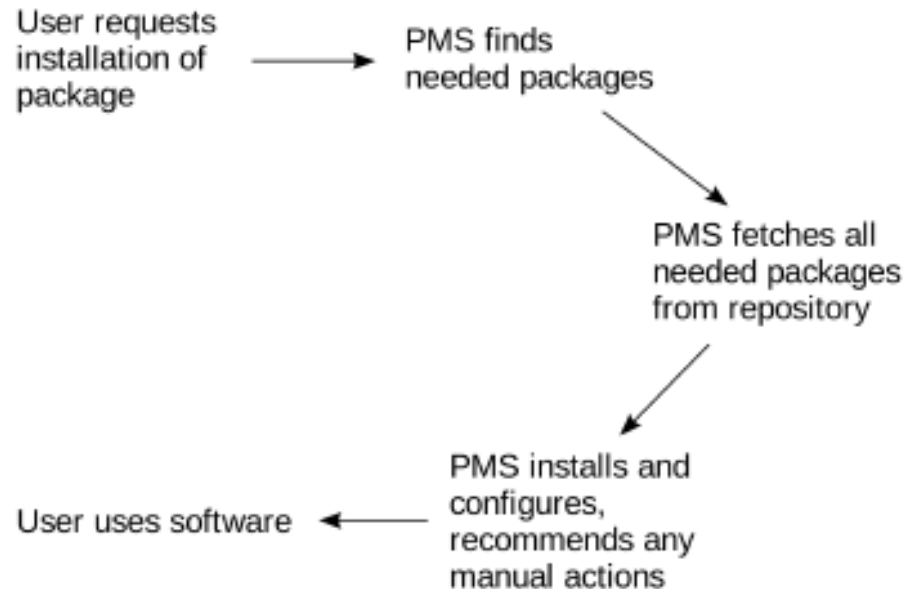
A package manager deals with packages, distributions of software and data in archive files. Packages contain metadata, such as the software's name, description of its purpose, version number, vendor, checksum, and a list of dependencies necessary for the software to run properly. Upon installation, metadata is stored in a local package database.



Package Management



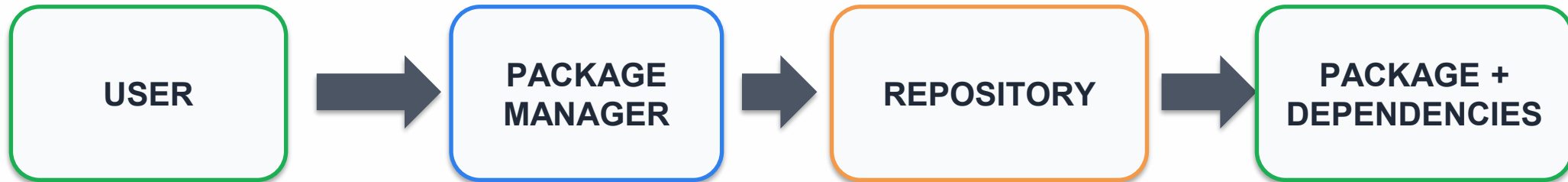
Package managers are designed to eliminate the need for manual installs and updates. This can be particularly useful for large enterprises whose operating systems are typically consisting of hundreds or even tens of thousands of distinct software packages.



How Linux Package Management Works



Package managers obtain software and metadata from configured repositories.



PACKAGE MANAGER

- Resolves dependencies
- Installs / updates / removes software

REPOSITORY

- Packages + metadata
- Updates and package signatures

Package Management



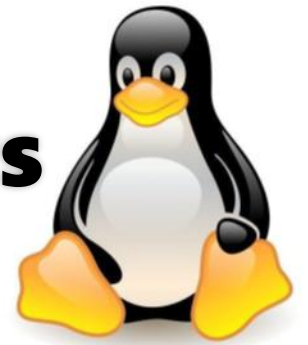
Operating System	Format	Tool(s)
Debian	.deb	apt, apt-cache, apt-get, dpkg
Ubuntu	.deb	apt, apt-cache, apt-get, dpkg
CentOS	.rpm	yum
Fedora	.rpm	dnf
FreeBSD	Ports, .txz	make, pkg



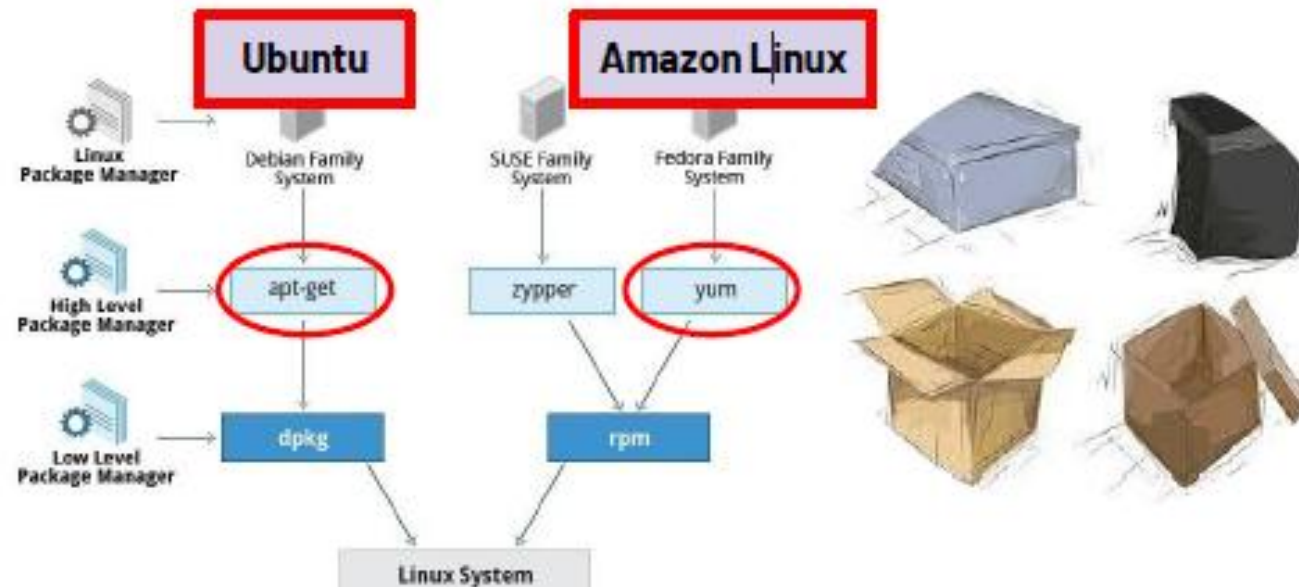
Popular Linux System Package Managers



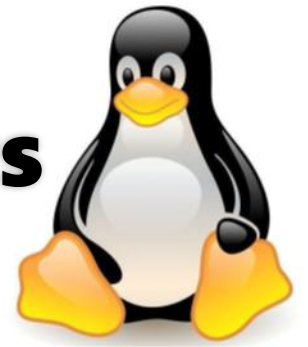
Popular Linux System Package Managers



Linux distributions use package managers to install, update, and remove software. Each distribution family normally uses its own supported package-management tools and repositories.



Popular Linux System Package Managers

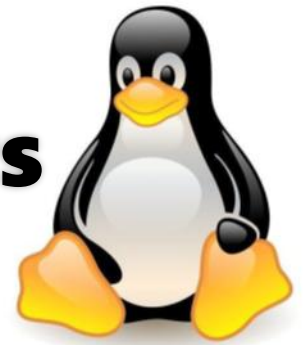


Debian Package Managers

dpkg is the main package management program for the Debian Linux distros. It is used to handle Debian package files with the extension of .deb

```
$ dpkg -i [package-name] # Installing a package
$ dpkg -r [package-name] # Removing a package
$ dpkg -l                 # Lists installed packages
```

Popular Linux System Package Managers



Debian Package Managers

APT (Advanced Package Tool)

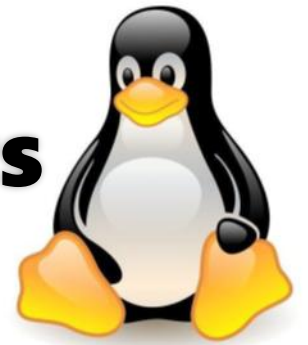
- The Advanced Packaging Tool is what Ubuntu Software Center is built on



- 'apt-get install PACKAGE' will install and organize software
- 'apt-cache list PACKAGE' will search for PACKAGE in the local database
- 'apt-get update' update the local package database

```
$ apt update           # Update the installed packages
$ apt install [package-name] # Install a package and all its dependencies
$ apt remove [package-name] # Remove a package
$ apt purge [package-name]  # Remove a package and its configuration files
```

Popular Linux System Package Managers



Debian Package Managers



Aptitude Package Manager

aptitude tool provides the functionality of **apt-get**, as well as many additional features:

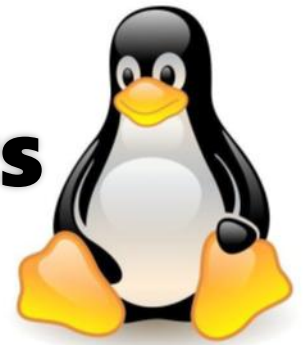
- aptitude provides easy access to all versions of a package
- aptitude tracks of obsolete software
- aptitude has a powerful system for searching particular packages

```
$ aptitude install [package-name] # Install a package
```

```
$ apt-get install [package-name] # Install a package
```



Popular Linux System Package Managers



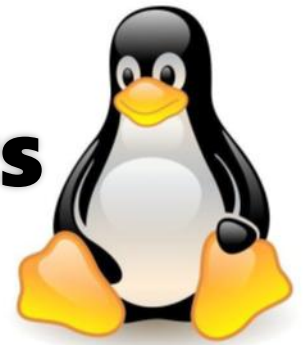
Red Hat Package Managers

rpm is the package manager for Red Hat Linux operating systems. The installation package files have .rpm extension. These files are used for installing programs. rpm command has been used for RPM packages by default but new tools are developed for better performance.

```
$ rpm -i [package-name] # Install a package  
$ rpm -e [package-name] # Uninstall a package
```



Popular Linux System Package Managers



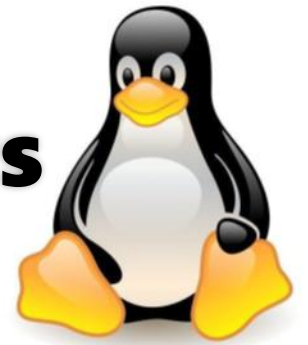
Red Hat Package Managers

YUM was the standard high-level package manager on many older RHEL and CentOS systems. On modern RHEL-family systems, DNF is the primary package manager. The yum command may still be available for compatibility.

```
$ yum install [package-name]    # Install a package
$ yum remove [package-name]    # Remove a package
$ yum update [package-name]    # Update a package
```



Popular Linux System Package Managers



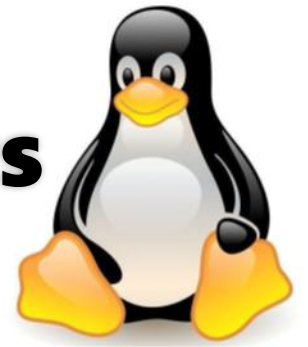
Red Hat Package Managers

➡ DNF – Dandified Yum

DNF is the modern high-level package manager used by Fedora, RHEL 8/9, and Amazon Linux 2023. It installs, updates, removes, and resolves package dependencies. Its command syntax is similar to YUM.

```
$ yum install dnf    # Install DNF via yum.  
$ dnf --version      # Checking DNF version  
$ dnf install        # Installing a package
```

Popular Linux System Package Managers



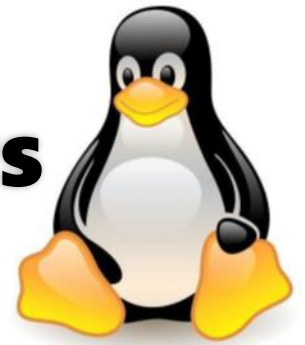
Red Hat Package Managers



Other RPM tools:

- zypper (openSUSE)
- up2date (Red Hat Enterprise Linux, CentOS 3 and 4, and Oracle Linux)
- urpmi (Mandriva Linux, ROSA Linux, and Mageia)
- apt-rpm (Ark Linux,[11] PCLinuxOS and ALT Linux)
- smart (Unity Linux and Fedora)
- rpmquery (Red Hat Enterprise Linux)

Popular Linux System Package Managers



Other Package Managers

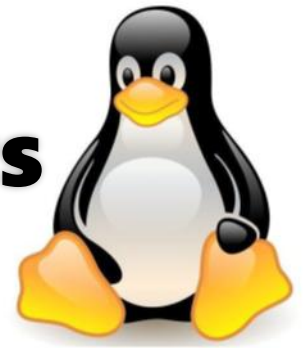


Below are a few more notable/interesting package managers.

- Portage: Package manager for Gentoo.
- Pacman: Arch Linux Package manager.
- Nix: A 'Fully Functional/Transactional' package manager.
- Brew: An Open Source package manager for OSX.**
- Chocolatey: A package manager for Windows.**



Popular Linux System Package Managers



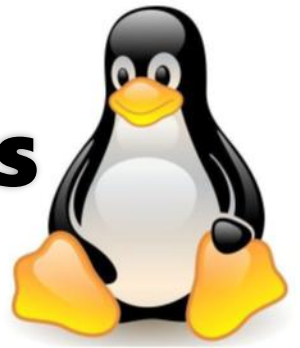
Other Package Managers

Programming languages have their own default package managers. They help to find and install the packages via searching libraries that exist on the internet for that language.

Examples: **Python: pip** / Ruby: gem, rubygems / Haskell: cabal / NodeJS: npm



Popular Linux System Package Managers



dnf install [package-name] → Install a package

dnf -y install [package-name] → Skip confirmation during installation

dnf remove [package-name] → Remove a package

dnf autoremove → Remove unused dependencies

dnf upgrade [package-name] → Upgrade a specific package

dnf upgrade → Upgrade installed packages

dnf info [package-name] → Show package information

dnf list installed → List installed packages

dnf list --showduplicates [package-name] → List available versions

Amazon Linux 2023 uses dnf by default; yum remains available as a compatibility command.

Exercise-1

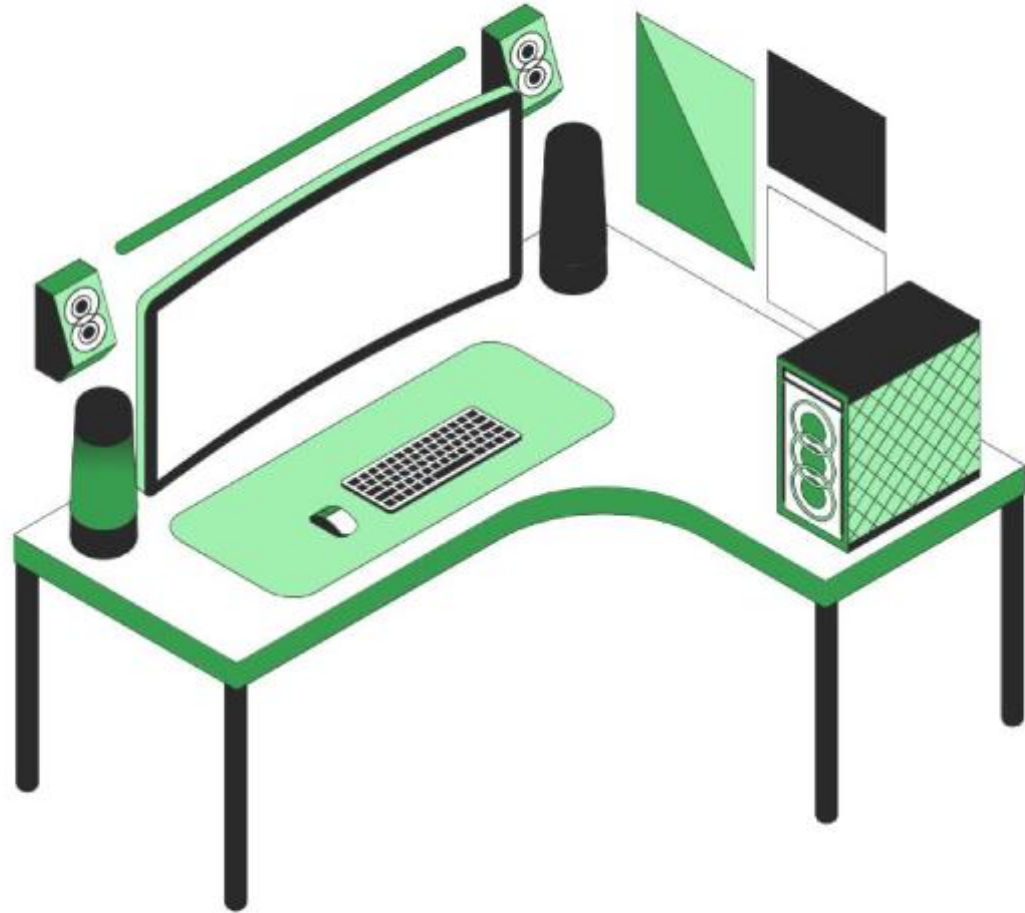


1. Update all installed packages
2. List all installed packages starting with "http"
3. Find all installed packages starting with "http"
4. If available, install httpd (skip confirmation during installation)
5. List the installed **httpd** package
6. Remove the **httpd** package
7. List the installed httpd package

Exercise-2



1. Remove Git along with unused dependencies
2. Check the installed Git version
3. Find the available previous Git version
4. Install the current previous Git version
5. Check the installed Git version
6. Update Git to the latest version
7. Check the installed Git version



Do you have any questions?

Send it to us! We hope you learned something new.